



FTF – Survey Invoicing & Billing (E2E)

AI Invoicing Agent — Client Documentation Package

Client: NexGen Surveying / FieldToFinish

Prepared by: Tisko Tech

Version: 1.0 · **Date:** 2026-07-16

Status: Draft for review

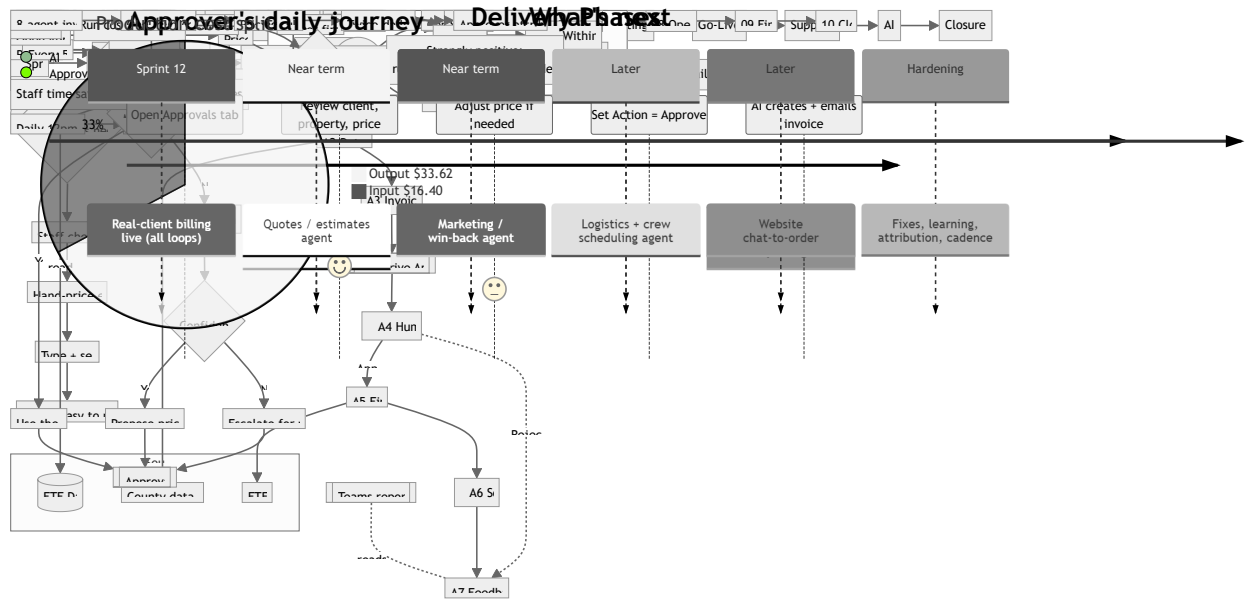
Document Index — FTF AI Invoicing Agent

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E)
Client	NexGen Surveying / FieldToFinish (FTF)
Prepared by	Tisko Tech
Project type	AI Agent / AI OS
Document	Master Document Index
Version	1.0
Owner	Prateek Chandra (CTO)
Status	Draft for review
Last updated	2026-07-16

Purpose

This is the map of the **whole repository** — the **client** package (`01_Client_Documents`) and the **internal** package (`02_Internal_Documents`). It tells you what each document is for and where to find it. Start here.

The package at a glance



Documents in this package

#	Folder	Document	What it answers
1	00_Project_Information	Project Profile	Who, what, why — the 1-page overview
2	00_Project_Information	Glossary	Plain-English meaning of every term
3	01_Client / 01_Discovery	Business Discovery & Requirements	The problem and what we agreed to build
4	01_Client / 02_Planning	Proposal, Solution & SOW	The solution, scope, and what's in / out
5	01_Client / 02_Planning	Project Roadmap & Solution Blueprint	The plan and how the system works
6	01_Client / 03_Project_Updates	Status & Milestone Report	Where the project stands now
7	01_Client / 04_Testing	UAT & Client Acceptance	How we prove it works + your sign-off
8	01_Client / 05_Go_Live	Go-Live Checklist & Handover	How we went live and what you now own
9	01_Client / 06_Support	Support & Maintenance Guide	How help works after go-live

#	Folder	Document	What it answers
10	01_Client / 06_Support	Training Guide & FAQ	How your team uses the system day to day
11	01_Client / 07_AI	AI Solution Guide	What the AI does, its limits, and safety
12	01_Client / 07_AI	AI Cost, Pricing & ROI	What it costs to run and the return
13	01_Client / 08_Project_Closure	Closeout & Future Roadmap	What was delivered + what's next

Internal package (02_Internal_Documents) — team-facing, not shared with the client

#	Folder	Document	What it answers
14	02 / 01_Project_Management	Project Charter, RAID & Decisions	Why, governance, risks, decisions, changes
15	02 / 02_Business_Analysis	Internal Requirements & Business Rules	Full requirements, pricing rules, stories, use cases
16	02 / 03_UI_UX	UX, Interface & Flows	Sheet / Teams / email surfaces + approver flow
17	02 / 04_Technical	Solution & Technical Architecture	Components, data, integrations, infra, security
18	02 / 05_AI_Engineering	AI Engineering Guide	Brain, prompts, guardrails, model, learning
19	02 / 06_Development	Development & Release Tracker	Sprint, modules, builds, releases (git-reconciled)
20	02 / 07_Testing	Test Plan, Cases & Results	Tests, regression, bugs, security/perf
21	02 / 08_Operations	Operations Runbook	Deploy, monitor, incident, backup, DR, SOPs
22	02 / 09_Finance	Finance, Cost & Profitability	Measured cost, margin, pricing, ROI (internal)
23	02 / 10_Project_Closure	Lessons Learned & Closeout	Retrospective + internal sign-off

Confidentiality: the internal set contains implementation detail (architecture, prompts strategy, infra paths, margin/pricing) and is **not** shared with the client. Neither package contains secrets, API keys, passwords, or client PII.

Reading order

- **Executives (client):** 1 → 4 → 6 → 12
- **Operations / approvers:** 10 → 11 → 9
- **Technical reviewers (internal):** 17 → 18 → 21
- **Delivery / PMO (internal):** 14 → 19 → 22 → 23

Revision history

Version	Date	Author	Change
1.0	2026-07-16	Tisko Tech	Initial client package index
1.1	2026-07-16	Tisko Tech	Added internal package (10 docs) to the index
1.2	2026-07-16	Tisko Tech	Multi-agent verification pass vs code + live prod server . Confirmed cadence <code>*/15</code> , caps A1 100 / A2 30 / A3 30, cost math, 8 agents, 6621 commits/0 tags. Corrected script names (<code>ftf-invoicing-*.sh</code> → <code>run_server_*.sh</code>), the editable breakdown column (col K , not G), GHA-disabled deployment note, and per-order cost (\$0.72). No secrets / no client-data leak.
1.3	2026-07-17	Tisko Tech	Cadence changed to <code>*/5</code> (every 5 min) across all docs. Added Col-F Regrid locality map link (opens the parcel locality, not global search) and the synced sheet backup (full workbook → server backups/ + OneDrive copy with an org share link, every cycle). Sheet Guide→v16 / How-To→v9.

Project Profile — FTF AI Invoicing Agent

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E)
Client	NexGen Surveying / FieldToFinish (FTF)
Prepared by	Tisko Tech
Document	Project Profile
Version	1.0 · Status Draft for review · Owner Prateek Chandra (CTO)
Last updated	2026-07-16

1. In one line

An **AI agent that prepares survey invoices automatically** and puts each one in front of a person to approve — so billing is faster, consistent, and never sent without a human "yes".

2. The problem it solves

Before	After
Staff hand-price and hand-type every invoice	AI drafts the invoice with a suggested price
Easy to miss orders that need billing	AI scans FieldToFinish every 5 minutes
Pricing varies by person	AI uses the same rules + learns from your edits
No single view of what's waiting	One "Approvals" sheet shows everything

3. What it is (project type)

An **AI Agent / AI OS** — a team of small AI workers (agents A0–A7) that run in sequence, 24/7, with a human approval gate in the middle.

4. How it works (30-second view)

- **Nothing is sent without human approval.** The AI only drafts and suggests.
- The human works in **one Excel "Approvals" sheet** on OneDrive.

- A **twice-daily Teams report** (12 PM & 7 PM ET) shows what happened and who approved what.

5. Key facts

Item	Detail
Build start	2026-05-20
Current phase	Live pilot on staging data, human-approval mode
Runtime	One production server, every 5 minutes, 24/7
Human interface	OneDrive Excel "Approvals" tab (+ Teams report)
AI provider	Anthropic Claude (Opus / Sonnet / Haiku)
Core systems	FieldToFinish API + database, Microsoft 365 (Excel + Teams)

6. People

Role	Who
Client sponsor	Ryan (NexGen)
Business owner / domain	Robert (NexGen Surveying)
Product / CTO	Prateek Chandra
Operations / approver	Sumit Parik, Prateek Chandra
Delivery team	Tisko Tech (AI, engineering, QA, DevOps)

[CLIENT INPUT REQUIRED] — confirm the official client legal name, logo, and the final list of named approvers for the cover pages.

7. Current status (traffic light)

Area	Status
Core pipeline (A0–A7)	● Built & running
Human approval flow	● Live (OneDrive sheet)
Monitoring / reporting	● Twice-daily Teams report
Production billing to real clients	● Gated — human-approval, staging-safe

Revision history

Version	Date	Author	Change
1.0	2026-07-16	Tisko Tech	Initial profile

Glossary — Plain English

Term	What it means
FTF / FieldToFinish	The survey order system where orders and invoices live
Order	One survey job for a property
Invoice	The bill sent to the client for a completed order
Approvals sheet	The OneDrive Excel tab where a person reviews and approves each draft
Agent (A0–A7)	A small AI worker that does one step of the job
A0 Orchestrator	The "conductor" that runs the other agents in order
A1 Flag Hunter	Finds orders that FTF has marked as needing an invoice
A2 Data Collector	Gathers the order's details (address, size, flood zone, etc.)
A3 Invoice Compiler	Drafts the invoice and suggests a price
A4 Human Gate	Reads your Approve / Reject / Hold decision
A5 Finalizer	Creates the real invoice in FTF once you approve
A6 Sender	Emails the invoice to the client
A7 Feedback Learner	Learns from your edits so future prices are better
<code>ng_invoice_needed</code>	The FTF flag that marks an order as "needs an invoice". The AI only brings flagged orders
Service / Breakdown by User	The blue column where you set what gets billed (Name: \$Amount)
Confidence	How sure the AI is about a price (High / Medium / Low)
Escalate	The AI flags an unusual order for a manager to review
Human-in-the-loop	A person must approve before anything is sent
Watermark	A "start from here" marker so old historical orders are not re-processed
Token	The unit AI providers bill by (roughly ¾ of a word)
Cron	A timer that runs the pipeline automatically (every 5 minutes)

Business Discovery & Requirements

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	Business Discovery & Requirements · Version 1.0 · Status Draft for review
Prepared by	Tisko Tech · Owner Prateek Chandra · Updated 2026-07-16

Purpose

Explain the business problem and list, in plain language, what the system must do.

Scope

Covers the **invoice / billing** part of the survey workflow: finding orders that need a bill, drafting the invoice, human approval, creating it in FTF, and emailing the client. Quoting/estimating and field scheduling are **out of scope** for this phase.

1. The business problem

Pain points

- Billing is manual and slow.
- Orders that need billing can be missed.
- Prices differ depending on who does them.
- No single place to see what is waiting.

2. Goals (what success looks like)

#	Goal	Measure
G1	Catch every order FTF flags for invoicing	100% of flagged orders reach the sheet
G2	Draft a priced invoice automatically	AI proposes price + line items
G3	Keep a human in control	Nothing sent without approval
G4	Make pricing consistent	Rules + learning from human edits

#	Goal	Measure
G5	Give one clear view + daily reporting	Approvals sheet + 2 Teams reports/day

3. Requirements

3.1 Functional (must do)

ID	Requirement	Status
FR-1	Scan FTF every 5 min for orders flagged <code>invoice_needed</code>	✓ Delivered
FR-2	Collect order details (address, size, county, flood zone, service)	✓ Delivered
FR-3	Draft invoice line items and suggest a price	✓ Delivered
FR-4	Detect condos, canceled, and delivered orders and flag them	✓ Delivered
FR-5	Post each draft to the OneDrive "Approvals" sheet	✓ Delivered
FR-6	Let a person Approve / Reject / Hold in the sheet	✓ Delivered
FR-7	Let a person edit the price/services before approving	✓ Delivered
FR-8	Create the invoice in FTF only after approval	✓ Delivered
FR-9	Email the invoice to the client	✓ Delivered
FR-10	Learn from human edits to improve future pricing	✓ Delivered
FR-11	Record who approved each order	✓ Delivered
FR-12	Twice-daily Teams report (activity + who did what)	✓ Delivered

3.2 Non-functional (how well)

ID	Requirement	Target
NFR-1	Safety: never send without human approval	Hard rule
NFR-2	No duplicate invoices	Idempotency guard on every create
NFR-3	Availability	24/7, self-heals each run
NFR-4	Traceability	Every action logged; approver captured
NFR-5	Privacy	Client data stays in FTF + Microsoft 365

4. In scope vs out of scope

✓ In scope	✗ Out of scope (this phase)
Invoicing flagged orders	Quotes / estimates
Human approval in Excel	Field crew scheduling
Pricing + learning	Marketing / win-back campaigns
Invoice email delivery	Website chat-to-order

5. Key business rules

- The AI **only** brings orders FTF flags `ng_invoice_needed = 1`.
- Condo / airspace units cannot be surveyed → auto-flagged, not billed.
- Canceled or already-invoiced orders are never re-billed.
- Prices come from the Pricing Rules tab first, then AI reasoning, then human edit.

6. Assumptions & dependencies

- FTF reliably sets the `invoice_needed` flag when an order should be billed.
- Microsoft 365 (Excel + Teams) is available to the approvers.
- Anthropic Claude API is available for pricing/extraction.

[CLIENT INPUT REQUIRED] — confirm whether unflagged, non-invoiced orders (e.g. quotes/in-progress) should ever be auto-surfaced. Today they are not, by design.

Approval

Name	Role	Decision	Date
	Client sponsor	☐ Approve ☐ Changes	

Revision history — 1.0 (2026-07-16, Tisko Tech): initial.

Proposal, Solution & Statement of Work (SOW)

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	Proposal, Solution & SOW · Version 1.0 · Status Draft for review
Prepared by	Tisko Tech · Owner Prateek Chandra · Updated 2026-07-16

Purpose

Describe the solution we proposed and built, and define the work scope, deliverables, and boundaries in one place.

1. The solution in plain terms

A **human-supervised AI billing assistant**. It watches FieldToFinish, prepares each invoice with a suggested price, and waits for a person to approve before anything is created or emailed.

2. What you get (deliverables)

#	Deliverable	Description
D1	Invoice pipeline (A0–A7)	The full 8-agent automated workflow
D2	Approvals sheet	One OneDrive Excel tab to review & approve
D3	Pricing rules + learning	Consistent pricing that improves over time
D4	Twice-daily Teams report	Activity + who approved what
D5	Guide & How-To tabs	Built-in instructions inside the workbook
D6	Documentation package	This client set (+ internal docs on request)

3. Scope

✓ Included	✗ Not included (this phase)
Invoicing FTF-flagged orders	Quotes / estimates automation
Data collection + AI pricing	Field scheduling / crew routing

✓ Included	✗ Not included (this phase)
Human approval workflow	Marketing / re-engagement agents
Invoice creation + email	Website chat-to-order
Monitoring + reporting	Accounts-receivable chasing

4. Approach & phases

5. Ways of working

- Short iterations; changes tracked as Change Requests.
- Human-approval safety kept at all times.
- All code version-controlled; deployed to one production server.

6. Acceptance

Work is accepted when the requirements in *Business Discovery & Requirements* are met and the client signs the **UAT & Client Acceptance** document.

7. Commercials

[CLIENT INPUT REQUIRED] — fees, payment schedule, and contract term are set in the signed commercial agreement and are intentionally **not** reproduced here. Running (usage) costs are summarized in **AI Cost, Pricing & ROI**.

8. Assumptions

- FTF flags orders for invoicing reliably.
- Client provides Microsoft 365 access for approvers.
- Production go-live for real-client billing is authorized separately by the client.

Approval

Name	Role	Decision	Date
	Client sponsor	☐ Approve ☐ Changes	

Project Roadmap & Solution Blueprint

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	Roadmap & Solution Blueprint · Version 1.0 · Status Draft for review
Prepared by	Tisko Tech · Owner Prateek Chandra · Updated 2026-07-16

Purpose

Show the delivery roadmap and how the solution is put together — enough for a business or technical reader to understand the design without jargon.

1. Roadmap

Phase	Outcome	Status
Sprint 0	Repo, infra, DB, 5 API connections	✔ Complete
Sprints 1–9	Detection, pricing, human gate, delivery, memory	✔ Complete
Sprint 10	Full staging test — GO	✔ Complete
Sprint 11	8-agent invoice pipeline + OneDrive approvals	✔ Complete
Now	Pilot on staging data + hardening	🟡 In progress
Sprint 12	All loops live for real-client billing	🟡 Pending client GO
Future	Quotes, marketing, logistics agents	🟡 Backlog

2. Solution blueprint (how it works)

3. What each agent does

Agent	Job	Human involved?
A0	Runs the others in order, every 5 min	No
A1	Finds orders FTF flagged for invoicing	No
A2	Collects address, size, county, flood zone, service	No
A3	Drafts line items + suggests price; flags condos/canceled/delivered	No
A4	Reads Approve / Reject / Hold from the sheet	Yes
A5	Creates the invoice in FTF (after approval)	No
A6	Emails the invoice to the client	No
A7	Learns from your edits for next time	No

4. The human touchpoint

The only place a person works is the **Approvals sheet**:

5. Technology (plain list)

- **AI:** Anthropic Claude (Opus / Sonnet / Haiku) for pricing and reading order data.
- **Human interface:** Microsoft 365 — Excel (OneDrive) + Teams.
- **Core system:** FieldToFinish API + database.
- **Runtime:** one always-on production server; timer every 5 minutes.

6. Safety & control (built in)

- No invoice is created or emailed without a human **Approve**.
- Duplicate-invoice guards on every create.
- Condo / canceled / delivered orders are flagged, not billed.

- Every action is logged; the approver's name is recorded.

Revision history — 1.0 (2026-07-16, Tisko Tech): initial.

Status & Milestone Report

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	Status & Milestone Report · Version 1.0 · Reporting date 2026-07-16
Prepared by	Tisko Tech · Owner Prateek Chandra

Purpose

A one-page snapshot of where the project stands. Reusable as a weekly template.

1. Overall health

KPI	Status
Scope	● On track
Schedule	● Core delivered; pilot ongoing
Quality	● Stable; issues fixed same-day
Risk	● Depends on FTF's invoice flagging
Production billing	● Gated on client GO

2. Milestones

Milestone	Target	Status
Foundation & connections	Sprint 0	✓ Done
Core agents (detect→deliver)	Sprints 1–9	✓ Done
Staging GO/NO-GO	Sprint 10	✓ GO
8-agent invoice pipeline + Approvals sheet	Sprint 11	✓ Done
Pilot on staging + hardening	Now	● In progress
All loops live for real clients	Sprint 12	■ Pending GO

3. Recent work (this period)

- Added 4 reference columns to the sheet (Property Size, Map Link, FEMA Zone, Service Type).
- Client Name now shows the **company** (ordering party), not the contact person.
- Fixed invoice line so the **service name you type is what appears** on the invoice.
- Strengthened learning from operator notes; applies to similar future orders.
- Intake raised to catch **all** new flagged orders; run cadence now **every 5 min**.
- Added a **twice-daily Teams report** (12 PM & 7 PM ET) with activity + who approved.
- Approvals now attributed to the **real person** (Sumit / Prateek), not a fixed name.

4. Numbers at a glance

Metric	Value	Source
Build window	2026-05-20 → 2026-07-14	git
Commits	6,621	git
Version tags	0 (releases tracked by feature commits)	git
Agents in pipeline	8 (A0–A7)	codebase
Run cadence	Every 5 min (pipeline) + 5 min (approvals)	prod cron

5. Risks & mitigations

Risk	Impact	Mitigation
FTF doesn't flag an order for invoicing	Order won't appear	Staff handle manually; option to broaden intake
Workbook open while system writes	Dropdown/format lag	Self-heals next run; close/reopen to refresh
Simultaneous approvers	"Who approved" may attribute to most-recent editor	Optional "Approved By" column for exactness

6. Next steps

- Continue pilot; collect operator feedback via the sheet.
- Confirm production-billing GO with the client (Sprint 12).

[CLIENT INPUT REQUIRED] — confirm the pilot end date and the go-live decision owner.

Revision history — 1.0 (2026-07-16, Tisko Tech): initial template + current status.

UAT & Client Acceptance

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	UAT & Client Acceptance · Version 1.0 · Status Draft for review
Prepared by	Tisko Tech · Owner Prateek Chandra · Updated 2026-07-16

Purpose

Show how we prove the system works, and give the client a simple sign-off.

1. How to test (5 steps)

2. Acceptance test cases

#	Scenario	Steps	Expected result	Pass?
UAT-1	New flagged order appears	Wait for a flagged order	Row appears with price + details	☐
UAT-2	Reference data shows	Open the row	Size, Map Link, FEMA Zone, Service filled	☐
UAT-3	Edit a price	Change Service / Breakdown by User	Total recalculates; name shows on invoice	☐
UAT-4	Approve	Set Action = Approve	Invoice created in FTF + emailed	☐
UAT-5	Reject	Set Action = Reject	No invoice, no email; AI learns	☐
UAT-6	Hold	Set Action = On-hold	Paused; can approve later	☐
UAT-7	Condo order	Wait for a condo	Auto-flagged, not billed	☐
UAT-8	No duplicate	Approve an already-invoiced order	System skips; no second invoice	☐
UAT-9	Who approved	Approve as a user	Report shows the real approver	☐
UAT-10	Daily report	Wait for 12 PM / 7 PM ET	Teams report with activity + by whom	☐

3. Entry & exit criteria

Entry (before UAT)	Exit (UAT done)
Pipeline running on staging	All critical cases pass
Approvers have sheet access	No open blocker defects
Test orders available	Client signs acceptance below

4. Defect handling

- Log any issue with a short note in the sheet or to Tisko Tech.
- Severity 1 (blocks billing) fixed same day; minor issues batched.

[CLIENT INPUT REQUIRED] — assign the UAT tester(s) and the acceptance date.

5. Client acceptance sign-off

Statement		Yes/No	
The system meets the agreed requirements		☐	
We accept it for the pilot / production stage indicated		☐	

Name	Role	Signature	Date
	Client sponsor		
	Operations lead		

Revision history — 1.0 (2026-07-16, Tisko Tech): initial.

Go-Live Checklist & Handover Guide

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	Go-Live Checklist & Handover · Version 1.0 · Status Draft for review
Prepared by	Tisko Tech · Owner Prateek Chandra · Updated 2026-07-16

Purpose

The checklist to go live safely, and a clear handover of what you now own.

1. Go-live readiness checklist

#	Item	Ready?
1	Pipeline runs on the production server every 5 min	☒
2	Approvals sheet live with Guide + How-To tabs	☒
3	Action dropdown (Approve/Reject/On-hold) working	☒
4	Duplicate-invoice guards active	☒
5	Twice-daily Teams report delivering	☒
6	Approver identity captured	☒
7	Approvers trained (see Training Guide)	☐
8	Client GO for real-client billing	☐
9	Support contact + hours agreed	☐

2. Go-live flow

3. What you now own (handover)

Item	You own	Notes
Approvals sheet	✓	Approve/edit in the blue columns
Pricing Rules tab	✓	Set fixed prices; no coding
Learning column	✓	Teach the AI in plain English
Daily Teams report	✓	Read-only monitoring
Server + pipeline	Tisko Tech (managed)	Support & maintenance

4. Rollback / stop

- To pause billing: leave Action blank (nothing is sent).
- To stop the whole pipeline: Tisko Tech disables the timer (5-minute action).
- No invoice is ever sent without a human Approve, so there is no "runaway" risk.

5. Handover contents

- This documentation package.
- Access to the Approvals workbook (Excel/OneDrive).
- The Teams report channel.
- Support contact (below).

[CLIENT INPUT REQUIRED] — confirm go-live date, first-week monitor owner, and the support contact + hours.

Sign-off to go live

Name	Role	Decision	Date
	Client sponsor	☐ GO ☐ HOLD	

Revision history — 1.0 (2026-07-16, Tisko Tech): initial.

Support & Maintenance Guide

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	Support & Maintenance Guide · Version 1.0 · Status Draft for review
Prepared by	Tisko Tech · Owner Prateek Chandra · Updated 2026-07-16

Purpose

Explain how help works after go-live and what keeps the system healthy.

1. Who to contact

Need	Contact
Something looks wrong on the sheet	Operations lead → Tisko Tech
Invoice not created / email not sent	Tisko Tech support
Pricing consistently wrong	Add a Pricing Rule, or tell Tisko Tech
Emergency (billing must stop)	Tisko Tech (pause the pipeline)

[CLIENT INPUT REQUIRED] — support email/phone, hours, and response times (SLA).

2. Support levels (suggested)

Severity	Example	Target response
S1 Critical	Wrong invoices could be sent	Immediate; pause pipeline
S2 High	Orders not appearing at all	Same business day
S3 Medium	A price is off	1–2 business days
S4 Low	Wording / cosmetic	Next release

3. What maintenance covers

- Keeping the pipeline running 24/7 on the server.
- Fixing defects and adjusting rules.
- Small enhancements within scope.
- Watching the daily report and cost.

4. Routine health

- The pipeline **self-heals** each run (a bad run doesn't block the next).
- A failure alert email is sent to the team automatically.
- The daily report surfaces anything stuck.

5. Simple things the client can do (no coding)

Task	How
Fix a price for a client/county	Add a row in the Pricing Rules tab
Teach the AI something	Type it in Learning provided by user
Pause an order	Leave Action blank or set On-hold
See what happened	Read the twice-daily Teams report

6. Backups & continuity

- Order + invoice data live in FieldToFinish (client system of record).
- Pipeline state is stored and version-controlled.
- The server can be rebuilt from code if needed.

[CLIENT INPUT REQUIRED] — confirm backup ownership and disaster-recovery expectations.

Revision history — 1.0 (2026-07-16, Tisko Tech): initial.

Training Guide & FAQ

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	Training Guide & FAQ · Version 1.0 · Status Draft for review
Prepared by	Tisko Tech · Owner Prateek Chandra · Updated 2026-07-16

Purpose

Teach an approver to use the system in a few minutes. No technical knowledge needed.

1. The only place you work

The **Approvals** tab in the OneDrive workbook. Blue columns are yours to edit; gray columns are managed by the AI.

2. Your daily routine (5 steps)

3. The service-entry standard

Type each service as **Service Name: \$Amount** , separated by a pipe | .

Example	Meaning
Boundary Survey: \$500.00	One service, \$500
Boundary Survey: \$500.00 Elevation Certificate: \$150.00	Two services, \$650 total

- The **name you type is the name that prints on the invoice**.
- Change the number to reprice; the total updates automatically.
- Then set **Action = Approve**.

4. The three actions

Action	What happens
Approve	Real invoice is created in FTF and emailed to the client

Action	What happens
Reject	Nothing is sent; the AI learns from it
On-hold	Paused; approve or reject later
(blank)	Ignored until you choose

5. Teach the AI (no coding)

- **One order:** type a note in **Learning provided by user** (e.g. "this client is always \$1,800 for a boundary"). The AI reads it next run and applies it to similar orders.
- **A permanent price:** add a row in the **Pricing Rules** tab.

6. FAQ

Question	Answer
Some order numbers are missing on the sheet — why?	The AI only brings orders FTF flags as <i>needing an invoice</i> . Quotes / in-progress orders aren't flagged, so they don't appear. This is normal.
An order I expected isn't here.	It likely isn't flagged <code>invoice_needed</code> in FTF yet. When FTF flags it, it appears on the next run.
Will it ever send without me?	No. Nothing is created or emailed without your Approve .
Can it send twice?	No. Duplicate guards stop a second invoice.
I don't see the Approve dropdown.	Close and reopen the workbook — your view was cached.
The dollar total didn't update.	Keep the <code>Name: \$Amount</code> format and the <code>\$</code> ; it recalculates each run.
Who approved an order?	The twice-daily Teams report shows the approver by name.
A condo order appeared.	It's auto-flagged as "cannot survey" and not billed — contact the client.

7. Getting help

Ask your operations lead, or contact Tisko Tech support (see Support & Maintenance Guide).

[CLIENT INPUT REQUIRED] — confirm who runs the 30-minute live training session and when.

Revision history — 1.0 (2026-07-16, Tisko Tech): initial.

AI Solution Guide

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	AI Solution Guide · Version 1.0 · Status Draft for review
Prepared by	Tisko Tech · Owner Prateek Chandra · Updated 2026-07-16

Purpose

Explain, for a non-technical reader, what the AI does, what it does **not** do, and the safeguards that keep it trustworthy.

1. What the AI is for

To **prepare** survey invoices — read the order, decide a fair price, and draft the bill — so a person only has to **check** and **approve** instead of building each one.

2. What the AI can do

Capability	Plain meaning
Read order data	Pulls address, lot size, county, flood zone, service type
Look at the property	Uses parcel/aerial + county data as pricing context
Suggest a price	Uses your rules + market history + reasoning
Explain itself	Gives a short reason and a confidence level
Detect special cases	Condo, canceled, delivered, duplicate orders
Learn	Adjusts from your edits and notes

3. What the AI will NOT do

Limit	Why it matters
Never sends without approval	You are always in control
Never invents an order or client	Only works from real FTF data
Won't price truly unusual jobs	It escalates to a manager instead
Won't re-bill an invoiced order	Duplicate guards
Won't survey condos	Auto-flags them as out of scope

4. How it decides a price

5. Confidence & human review

Confidence	Meaning	Suggested action
● High	Strong data + clear rule/history	Quick check, approve
● Medium	Reasonable but verify	Review price
● Low / Escalate	Unusual or thin data	Manager reviews

Human review points: every order (approval gate), plus any Escalate flag.

6. Safety & guardrails

- **Approval gate:** nothing created/sent without a human Approve.
- **Duplicate prevention:** re-checks FTF before creating.
- **Canceled/condo/delivered guards:** flagged, never billed.
- **Learning is additive:** operator notes guide the AI; they don't bypass approval.
- **Full audit:** every step logged; the approver's name recorded.

7. Learning loop

8. Privacy & data

- Works only with data already in **FieldToFinish** and **Microsoft 365**.
- No client data is stored outside those systems and the project's own state.
- AI provider: **Anthropic Claude**; prompts contain order details needed to price.

[CLIENT INPUT REQUIRED] — confirm any data-handling / compliance clauses required in your client contracts (e.g. PII handling, retention).

9. Expected accuracy

- On standard orders with a rule or clear history: high.
 - Accuracy improves as the AI learns from approvals/edits.
 - Unusual jobs are escalated rather than guessed — by design.
-

Revision history — 1.0 (2026-07-16, Tisko Tech): initial.

AI Cost, Pricing & ROI

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	AI Cost, Pricing & ROI · Version 1.0 · Status Draft for review
Prepared by	Tisko Tech · Owner Prateek Chandra · Updated 2026-07-16
Measurement window	2026-05-20 → 2026-07-17 (~1.9 months)

Purpose

Show what the AI costs — using **measured** numbers, not estimates. Two costs are reported: what it costs to **run** (production) and the value of what it took to **build** (development).

1. The answer first

Cost	Amount	What it means
Production (run cost)	≈ \$50 total (~ \$26 / month)	The live AI spend to date
Development (build value)	≈ \$386 (measured local sessions)	List-rate value of build tokens on record
Cost per order (run)	< \$1 (~\$0.72)	AI cost to prepare one order

The run cost is **tiny** relative to the value of each survey invoice. One approved invoice typically covers **months** of AI run cost.

2. Production (run) cost — MEASURED, authoritative

Source: Anthropic **Admin Usage API**, filtered to this project's live key **FTF-Invoicing Agent** , priced at official list rates. Single model in production: **Claude Sonnet 4.6**. No cached tokens used.

Item	Tokens	Rate /MTok	Cost
Input	5,467,559	\$3	\$16.40
Output	2,241,239	\$15	\$33.62
Total			\$50.02

- **Monthly run rate:** ≈ **\$26 / month**.
- **Per order:** ≈ **\$0.72** (≈ \$50 ÷ ~69 orders currently tracked; illustrative).
- Confidence: **High** — isolated to the one production key and priced at list rates.

3. Development (build) cost — MEASURED (local sessions)

Source: this project's **Claude Code transcripts** on record, **deduped by message id**, model **Claude Opus 4.8**, priced at list rates.

Item	Tokens	Rate /MTok	Cost
Input	13,300	\$15	\$0.20
Output	629,125	\$75	\$47.18
Cache write	8,680,345	\$18.75	\$162.76
Cache read	117,063,738	\$1.50	\$175.60
Total			\$385.74

 **Scope note (honest):** only the **locally-retained** transcript is available (one session file). This figure is exact **for the sessions on record**, but it does **not** capture the full multi-month build across all machines/sessions — so true total build cost is **higher**. Development cost is a list-rate **value**, not an invoice.

4. ROI

Item	Before (manual)	After (AI + approve)
Prepare an invoice	several minutes each	seconds to draft + a quick check
Orders missed for billing	occasional	~0 flagged orders missed
Pricing consistency	varies by person	rule-based + learning
Monthly AI run cost	—	≈ \$26

[CLIENT INPUT REQUIRED] — provide current **staff minutes per invoice** and **invoices/month** to convert time

saved into a dollar ROI and a payback period.

5. How these numbers were produced (provenance)

- **Production:** `GET /v1/organizations/usage_report/messages` filtered by `api_key_ids=[FTF-Invoicing Agent]` , summed tokens, priced at list rates.
- **Development:** local `~/ .claude/projects/<this-project>/*.jsonl` , deduped by `message.id` , priced at list rates.
- Rates: Sonnet 4.6 \$3/\$15 per MTok; Opus 4.8 \$15/\$75, cache-write \$18.75, cache-read \$1.50.
- No figure is estimated; scope is strictly this project's key/transcripts.

Revision history

Version	Date	Author	Change
0.9	2026-07-16	Tisko Tech	Structure + method (cost pending)
1.0	2026-07-16	Tisko Tech	Filled with measured production + dev cost

Project Closeout & Future Roadmap

Field	Value
Project	FTF – Survey Invoicing & Billing Delivery (E2E) · Client NexGen Surveying / FTF
Document	Closeout & Future Roadmap · Version 1.0 · Status Draft for review
Prepared by	Tisko Tech · Owner Prateek Chandra · Updated 2026-07-16

Purpose

Summarize what was delivered in this phase and lay out what comes next.

1. What was delivered

Delivered	State
A0–A7 invoice pipeline (find → draft → approve → create → email → learn)	✓
OneDrive Approvals sheet + Guide/How-To tabs	✓
Reference columns (size, map link, FEMA zone, service type)	✓
Consistent pricing (rules) + learning from edits	✓
Twice-daily Teams monitoring report	✓
Human-approval safety + duplicate/condo/cancel guards	✓
Real approver captured per decision	✓

2. Lessons learned (kept short)

What we learned	Action taken
Excel Online caches an open file	Self-heal on next run; guide users to reopen
Approver identity isn't per-cell in Graph	Attribute via activity feed; optional exact column
The AI is only as complete as FTF's flag	Documented; option to broaden intake

3. Future roadmap

Item	Value	Priority
Production billing for real clients	Full automation benefit	★ Next
Quotes / estimates automation	Faster front-of-funnel	High
Marketing / win-back agent	Recover inactive clients	Medium
Logistics / crew scheduling	Operational efficiency	Later
Website chat-to-order	New order capture	Later

(Roadmap items are drawn from the project backlog; scope/pricing per future SOWs.)

4. Closeout checklist

Item	Done?
Deliverables accepted (UAT signed)	☐
Documentation handed over	☐
Support model agreed	☐
Future phases scoped	☐

[CLIENT INPUT REQUIRED] — confirm phase closeout date and sign-off owner.

Sign-off

Name	Role	Decision	Date
	Client sponsor	☐ Accept	

Revision history — 1.0 (2026-07-16, Tisko Tech): initial.